EXPERIMENT 7

Connect React Frontend with Express and MongoDB Backend

AIM

To develop a full-stack task application by connecting a React frontend with an Express and MongoDB backend.

ALGORITHM

1. Create a MongoDB database and connect it using Mongoose.
2. Define a task schema and Express API routes.
3. Create a React form to enter task details.
4. Send task data from React to Express using fetch().
5. Store the data in MongoDB and display the response.

PROGRAM

```
// Backend: server.js
// Install: npm install express mongoose cors
const express=require('express'), mongoose=require('mongoose');
const cors=require('cors'), app=express();
app.use(cors()); app.use(express.json());
mongoose.connect('mongodb://127.0.0.1:27017/taskdb');

const Task=mongoose.model('Task',new mongoose.Schema({title:String}));
app.get('/tasks',async(req,res)=>res.json(await Task.find()));
app.post('/tasks',async(req,res)=>{
 const task=await Task.create({title:req.body.title});
 res.status(201).json(task);
});
app.delete('/tasks/:id',async(req,res)=>{
 await Task.findByIdAndDelete(req.params.id); res.sendStatus(204);
});
app.listen(5000,()=>console.log('API running'));

// React component: App.jsx
import {useEffect,useState} from 'react';
export default function App(){
 const [tasks,setTasks]=useState([]),[title,setTitle]=useState('');
 const API='http://localhost:5000/tasks';
 const load=()=>fetch(API).then(r=>r.json()).then(setTasks);
 useEffect(()=>{load()},[]);
 async function add(){
  await fetch(API,{method:'POST',headers:{'Content-Type':'application/json'},
   body:JSON.stringify({title})});
  setTitle(''); load();
 }
 return <div><h2>Task Manager</h2>
  <input value={title} onChange={e=>setTitle(e.target.value)}/>
  <button onClick={add}>Add</button>
  {tasks.map(t=><p key={t._id}>{t.title}</p>)}
 </div>;
}
```

RESULT

Thus, the React frontend was connected successfully to an Express backend with MongoDB for task storage.

VIVA QUESTIONS

1. What is MongoDB?
2. What is Mongoose?
3. Why is CORS enabled?
4. How does React communicate with Express?
5. What is the purpose of useEffect()?